

1(a)	<i>idea of:</i> mix (equal volumes of), <i>Chlorella</i> / suspension, and sodium alginate ; (slowly) drip mixture into calcium chloride (solution) ; using a, syringe / pipette ;	3
1(b)(i)	<i>independent variable:</i> temperature ; <i>dependent variable:</i> distance travelled by the coloured liquid ;	2
1(b)(ii)	<i>any 2 from:</i> <i>idea that</i> 1 (suitable because) these are temperatures at which, enzymes are expected to be functioning / <i>Chlorella</i> may survive ; 2 (suitable because) gives enough range for at least five temperatures ; 3 (not suitable because) does not show effect of very, high / low, temperatures ;	2

Question	Answer	Marks
1(b)(iii)	<i>allow in context of using the same respirometer for each temperature, or separate respirometers set up in the same way for each temperature</i> <i>any 7 from:</i> 1 <i>idea of</i> adding a fixed volume of water to respirometer ; 2 <i>idea of</i> adding, fixed number / same mass, of immobilised algal balls (to the water below the shelf with CO₂ absorbent) ; 3 <i>idea of</i> attaching the tap, capillary tubing and scale so that they are airtight ; 4 <i>idea of</i> opening the tap (using syringe) to add air ; 5 <i>idea of</i> dipping capillary tube into liquid / setting position of liquid (using syringe) ; 6 <i>idea of</i> covering respirometer tube (containing <i>Chlorella</i>) with, metal foil / light proof material ; A carry out in a dark room 7 <i>idea of</i> placing respirometer tube into a water-bath at a known temperature ; 8 <i>idea of</i> opening tap and leaving to equilibrate ; 9 <i>idea of</i> closing the tap and marking position of liquid ; 10 <i>idea of</i> measuring movement of liquid along the scale for 5 minutes ; 11 <i>idea of</i> doing minimum of three measurements (in sequence) <u>and</u> taking a mean ; 12 <i>idea of</i> repeating experiment at each temperature, e.g. 20 °C, 30 °C, 40 °C and 50 °C ; A temperatures within the range of 10°C to 50°C 13 low risk experiment or medium risk experiment due to (named) CO₂ absorbent harmful <u>and</u> wear gloves ; 14 AVP ;	7
1(c)(i)	<i>idea of:</i> find radius of tubing in mm and use πr^2 (to find area) ; multiply by distance moved in mm ; A in terms of a formula	2

Question	Answer	Marks
1(c)(ii)	<i>any 2 from:</i> 1 <i>idea that</i> oxygen uptake increases as temperature increases <u>and</u> because enzymes, gain more kinetic energy / collide more frequently / form more enzyme-substrate complexes ; 2 <i>idea that</i> increases faster up to 30°C and then slows down <u>and</u> because, optimum temperature exceeded / other factors become limiting ; A doubles every 10°C increase up to 30°C then slows / Q₁₀ plus explanation 3 AVP ; e.g. <i>idea that</i> increase in metabolic rate is due to increasing enzyme activity	2